📡 Amateur Radio Basic Course (10 Weeks, 2 Hours Each)

Goal:
Get you to **80% or higher** on the Basic exam, with real understanding (not just memorization).

Week 1 — What This Is + How Radios Work (Big Picture First)

What you'll learn

- What amateur radio actually is (and isn't)
- The full **radio system in one pass**
- How voice/data gets from one person to another

What we'll do

- Build the **6-step radio chain** on the board
- Walk through a real signal path (mic → speaker)
- Short intro quiz (baseline)

Why this matters
This is your mental "map." Everything else plugs into this.

Study Guide

- Chapter 1 (Introduction)
- Chapter 13 (just the idea of modulation — light)

Question Bank

- Basic definitions and service purpose (B-001 section)

Week 2 — Electricity Basics (Only What You Need)

What you'll learn

- Voltage, current, resistance
- Ohm's Law (simple use, not heavy math)
- Power (watts)

What we'll do

- Solve a few practical problems
- Predict what happens when values change
- Quick quiz at end

Why this matters
All radio equipment runs on these ideas.

Study Guide

- Chapter 2 (Basics)
- Chapter 3 (Ohm's Law and Power)

Question Bank

- Electrical fundamentals (B-002, B-003 types)

Week 3 — AC, Frequency, and Waves

What you'll learn

- AC vs DC
- Frequency and wavelength
- What a radio wave actually is

What we'll do

- Work simple wavelength calculations
- Relate frequency to amateur bands
- Visual wave sketches

Why this matters
Radio = fast-changing electricity turned into waves.

Study Guide

- Chapter 2 (AC sections)
- Chapter 5 (Waves, Frequency, Bands)

Question Bank

- Frequency, wavelength, band questions

Week 4 — Active Devices (Diodes, Transistors, Tubes)

What you'll learn

- What diodes do (rectify, detect)
- What transistors do (amplify, switch)
- Basic idea of tubes

What we'll do

- Tie each device to a job in the radio
- Identify where they appear in transmitters/receivers
- Short quiz

Why this matters
These are what make radios actually work.

Study Guide

- Chapter 9 (Active Devices)

Question Bank

- Device function and behavior questions

Week 5 — How Radios Work (Now in Detail)

What you'll learn

- Transmitter and receiver blocks
- Modulation (AM, FM, SSB, CW — concept level)
- Signal flow through a radio

What we'll do

- Rebuild the radio system with detail
- Trace signals through blocks
- Practice identifying functions in exam-style questions

Why this matters
This ties together Weeks 1–4 into a working system.

Study Guide

- Chapter 13 (Modulation and Transmitters)
- Chapter 14 (Receivers)

Question Bank

- Modulation and receiver/transmitter questions

Week 6 — Antennas (High-Value Topic)

What you'll learn

- What antennas actually do
- Basic antenna types (dipole, vertical, beam)
- Resonance and length (simple formulas)

What we'll do

- Sketch antenna patterns
- Do simple length calculations
- Identify antennas in questions

Why this matters
This is a **high-scoring exam area**.

Study Guide

- Chapter 8 (Antennas)

Question Bank

- Antenna types, behavior, length

Week 7 — Propagation (Why Signals Travel)

What you'll learn

- How signals travel (HF vs VHF/UHF)
- Ionosphere basics
- Why signals go far (or don't)

What we'll do

- Real-world examples (day vs night, HF skip)
- Match propagation to bands
- Quick quiz

Why this matters
Explains "mystery" behavior on the air.

Study Guide

- Chapter 6 (Propagation)

Question Bank

- Propagation and band behavior questions

Week 8 — Feedlines, SWR, and Practical Issues

What you'll learn

- Coax vs balanced lines
- SWR (what it means, not heavy math)
- Losses and matching (concept only)

What we'll do

- Diagnose simple "station problems"
- Interpret SWR questions
- Practice questions

Why this matters
Common exam topic + real-world operation.

Study Guide

- Chapter 7 (Transmission Lines)
- Chapter 8 (SWR sections)

Question Bank

- SWR, feedline, impedance basics

Week 9 — Rules, Operating, and Safety

What you'll learn

- Canadian regulations
- Call signs, band limits, power limits
- Operating basics and etiquette
- Safety (electrical, RF, antennas)

What we'll do

- Work through typical regulation questions
- Practice identifying correct/incorrect operation
- Timed quiz

Why this matters
This is a **large portion of the exam**.

Study Guide

- Chapter 12 (Operation)
- Chapter 16 (Safety)
- Chapter 17 (Regulations)
- RBR-4 document

Question Bank

- B-001 (regulations-heavy section)

Week 10 — Exam Prep and Practice

What you'll do

- Full-length practice exam
- Review weak areas
- Strategy for test day

Focus

- Recognizing question patterns
- Avoiding common mistakes
- Time management

Why this matters
This is where most students either:

- lock in 80%+, or
- realize what they missed

Study Guide

- Review all relevant chapters

Question Bank

- Mixed questions across all sections

What You Should Expect Overall

- **Every class includes questions** — not just lectures
- You'll see exam-style questions early and often
- Concepts are taught in the order that makes them stick
- The goal is **understanding first, then speed**

Straight talk

If you:

- show up
- do the weekly questions
- and take Week 10 seriously

You should be in the **80–90% range**, based on how the exam is structured and what it tests.

That's not a guess — it's based on how closely this sequence matches the actual question bank and study guide structure.